

Neural Architectures for Tweet Sentiment Analysis : A Comparative Study

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Abstract - This project investigates the trade-offs between model architecture and data scale in performing sentiment analysis on Twitter data. We systematically benchmarked four distinct neural architectures : Multilayer Perceptrons (MLP), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and Transformer-based Encoders. These models were evaluated across both limited (200k) and large-scale (2.5M) datasets. Our results demonstrate that increasing the amount of training data generally leads to improved model performance ; however, more advanced and complex architectures are better able to effectively leverage larger datasets, allowing them to extract richer representations and achieve greater performance gains than simpler models. Notably, our 1D CNN implementation achieved a validation accuracy of 85.77% on the full dataset, performing competitively with the more computationally intensive LSTM (86.31%). These findings suggest that for short, informal text, lightweight and parallelizable architectures like CNNs offer an optimal balance of training efficiency and predictive accuracy when leveraged with sufficient data.

1 Introduction

Sentiment analysis is a core task in natural language processing, with applications in social media monitoring, opinion mining, and market analysis. Tweets present a particularly challenging setting for sentiment classification due to their short length, informal language, frequent use of slang and emojis, and the absence of consistent grammatical structure. As a result, both the choice of model architecture and the representation of textual input play a critical role in determining performance.

In this project, we study tweet-level sentiment analysis by comparing a range of neural network architectures with increasing modeling capacity. We begin with a multilayer perceptron (MLP) as a non-sequential baseline, and progressively evaluate convolutional neural networks (CNNs), recurrent neural networks based on Long Short-Term Memory (LSTM), and transformer-based models. These architectures dif-

fer in their ability to capture word order, local context, and long-range dependencies, making them well suited for analyzing how architectural choices impact performance on short informal text.

The main contributions of this work are threefold. First, we provide a systematic comparison of neural architectures for tweet sentiment analysis under controlled experimental settings. Second, we quantify the impact of embedding dimensionality and dataset size on model performance. Third, we analyze the strengths and limitations of each model, highlighting the trade-offs between simplicity, expressiveness, and scalability. Our results offer practical insights into selecting appropriate models for sentiment analysis on short social media text.

2 Data

We address a binary tweet sentiment classification task, where each tweet is labeled according to whether it originally contained a positive or negative smiley. All smileys have been removed from the text, requiring the model to infer sentiment implicitly from the remaining words. The resulting dataset contains informal and noisy Twitter language, including slang, abbreviations, and emojis.

To represent textual input, we use word embeddings trained with the GloVe algorithm. The embeddings are constructed by computing word co-occurrence statistics over the training corpus and learning dense vector representations through matrix factorization. GloVe embeddings capture semantic relationships between words and provide a continuous representation that is well suited for neural models.

We use the default embedding dimensionality of 20 as a baseline and additionally train 100-dimensional embeddings to increase the expressive capacity of the embedding space. These embeddings are used as the input representation for all models evaluated in this work.

To analyze the effect of data scale, experiments are conducted on two training set sizes : a smaller subset of 200,000 tweets and the full dataset containing 2.5 million tweets. Tweets are short on average, with

most containing fewer than 30 tokens, making them suitable for sequence-based models.

3 Methods

3.1 MLP

A multilayer perceptron (MLP) is used as a baseline model to benchmark performance against sequence-based architectures. Unlike other models, the MLP operates on fixed-size input representations and does not explicitly model word order or temporal dependencies, making it representative of non-sequential approaches to text classification.

We evaluate all models in this study on four different configurations, by varying the training set size (200k and 2.5M tweets) and the embedding dimensionality (20 and 100). Table 1 summarizes the validation performance across all configurations.

TABLE 1 – MLP benchmark performance.

Training set	Emb. dim.	Best val. acc.
200k	20	0.7314
200k	100	0.7900
2.5M	20	0.7321
2.5M	100	0.7995

Increasing the embedding dimensionality leads to noticeable performance improvements, indicating that richer input representations benefit the MLP. In contrast, scaling the dataset from 200k to 2.5M samples yields only marginal gains, suggesting that the model is capacity-limited rather than data-limited. Even at the largest scale, the MLP remains significantly below the performance achieved by the LSTM.

These results highlight a key limitation of the MLP for sentiment analysis : the inability to capture word order and contextual dependencies. While the MLP serves as a fast, stable, and interpretable benchmark, its lack of sequence modeling fundamentally restricts its effectiveness for tweet-level sentiment classification.

3.2 CNN

Convolutional Neural Networks (CNNs), while originally designated for computer vision, have proven highly effective for text classification tasks. Unlike Recurrent Neural Networks which process text sequentially, CNNs utilize 1D convolution filters to slide over the input sequence, capturing local correlations between adjacent words. In the context of sentiment analysis, this allows the model to detect specific N-gram patterns (such as "not good" or "very happy") regardless of their position in the tweet. The architecture consists of an embedding layer followed by a 1D convolutional layer, a ReLU activation functions, and global max pooling. The max pooling operation extracts the most salient feature from the entire sequence, effectively allowing the model to focus on the strong sentiment indicator present in the tweet. This feature vector is then passed through a dropout layer for regularization and a final fully connected layer with a sigmoid activation for binary classification. A key implementation challenge was handling the memory constraints of the full 2.5M dataset ; this was addressed by implementing a lazy-loading dataset class that processes text sequences on-the-fly, enabling efficient training on consumer-grade hardware. We evaluated the CNN under the same four experimental configurations as the MLP and LSTM models. The results are summarized in Table 2.

TABLE 2 – CNN performance across dataset sizes and embedding dimensions.

Training set	Emb. dim.	Best val. acc.
200k	20	0.8253
200k	100	0.8237
2.5M	20	0.8598
2.5M	100	0.8625

The results demonstrate that the CNN significantly outperforms the baseline MLP, confirming the importance of capturing local context in sentiment analysis. While the CNN achieves accuracy competitive with the LSTM (0.8631 on the full dataset with 100-dimensional embeddings), it offers the distinct advantage of parallelizability, resulting in faster training times compared to the sequential processing required by recurrent models. The transition from the small

dataset (200k) to the full dataset (2.5M) provided a substantial boost in performance, validating the model’s ability to scale with increased data availability.

3.3 LSTM

Recurrent neural networks, and in particular Long Short-Term Memory (LSTM) models, are well suited for tweet-level sentiment analysis due to their ability to capture sequential dependencies in short text. Tweets typically contain a limited number of tokens, and sentiment cues may appear anywhere in the sequence, making order-sensitive models preferable to bag-of-words approaches.

We have employed an LSTM architecture to model tweet sequences using learned word embeddings. The observed tweet-length distribution indicates that most samples are shorter than 30 tokens, which matches the strengths of LSTMs and enables effective sequence modeling without requiring deep architectures or aggressive truncation. To study the impact of data scale and representation capacity, we evaluate four configurations obtained by varying the training set size and embedding dimensionality.

TABLE 3 – LSTM variations and validation performance.

Training set	Emb. dim.	Best val. acc.
200k	20	0.7940
2.5M	20	0.8429
200k	100	0.8335
2.5M	100	0.8631

Increasing the training set size yields the largest improvement : scaling from 200k to 2.5M tweets increases validation accuracy from 0.7940 to 0.8429 with 20-dimensional embeddings. This trend persists with larger embeddings, where accuracy improves from 0.8335 to 0.8631. Increasing the embedding dimensionality provides consistent gains by enabling richer semantic representations ; however, these gains are smaller than those obtained from data scaling, indicating diminishing returns with increased model capacity. Across all configurations, training and validation accuracies remain close, suggesting limited overfitting and good generalization.

3.4 Encoder based Classifier

Transformer architectures have become very popular in recent years with massive breakthroughs, notably in natural language processing tasks. Encoder only based models (like BERT and ViT) are very effective at extracting a latent representation of tokens based on the sequence context. This is due to the use of transformer blocks with self-attention, enabling the model to capture long range dependencies between tokens while allowing for better parallelization than recurrence based models.

The implemented architecture consists of three main parts : position encoding, multiple self-attention encoder layers, and a final classification layer. As word order is very important for text classification and self-attention does not account for the token order of the input, a position encoding layer is necessary to add positional information directly to the tokens themselves. The goal of the encoder layers is to embed the sentiment of the input sequence into one token that is then given to the classifier. The classification layer takes a single embedded token as input and, as the task is binary classification, outputs a single value.

The classification token can be an extra learnable parameter added to the input sequence or an arbitrary token of the sequence itself. The accuracy of both methods was compared on a smaller dataset of 200k samples with an embedding dimension of 20. The model with an extra parameter performed worse on multiple tests (best validation accuracy of 0.8181) than the model using the first input token of each sequence for classification (best validation accuracy of 0.8233). This may come as a surprise as classifying on a special token is very common in encoder based classification models. The most probable explanation for this discrepancy is a limited number of encoder layers in our model, due to computation limitations. The reduced number of encoder constrains the quality of the learned embeddings, consequently, the model achieves better performance when using a token with prior semantic meaning for classification than when required to learn the representation from the ground up. We thus chose the model performing classification on the first sequence token to evaluate four combinations of training set size and embedding dimensionality.

TABLE 4 – Encoder based classifier performance across dataset sizes and embedding dimensions.

Training set	Emb. dim.	Best val. acc.
200k	20	0.8233
200k	100	0.7998
2.5M	20	0.8321
2.5M	100	0.8628

Although the encoder based model performs significantly better than the baseline MLP model, the results do not show a performance gain compared to the LSTM and CNN models studied in this work. This is contrary to expectations, as transformer architectures have been shown to outperform recurrent and convolutional models on natural language processing tasks in recent years. Moreover, models such as BERT have been shown to achieve over 90% accuracy [1] when finetuned on the task of sentiment analysis. The lack of improvement over other models in this study is probably due to hardware limitations. This model is substantially larger than the LSTM and CNN models discussed above. On the full dataset with embedding dimension of 100, the training takes 20 minutes per epoch. Thus, we had to set the number of encoder layers to 6, limiting the models capacity to learn contextual representation of the tokens. The full training time of the final model required several hours on a GPU, which hindered experimentation with different hyperparameters and configurations.

Another noteworthy result is the accuracy of the model trained on the small dataset with embedding dimension of 100. This accuracy is notably lower than the configuration with embedding dimension of 20. This is also the only configuration that exhibited a large difference between accuracy on the training set (0.8787) and accuracy on the validation set, indicating a possible overfitting to the training data. The same model, trained on the large dataset, showed a negligible gap between training and validation accuracies, further suggesting the presence of overfitting on the small dataset.

4 Ethical Risks

An important ethical risk in tweet sentiment analysis is **biased performance across different ways of writing**. Tweets often contain slang, informal spelling, emojis, sarcasm, and identity-related terms. A sentiment model trained on social media data may learn incorrect associations between some of these expressions and sentiment labels, which can lead to systematic misclassification. The main stakeholders affected are (i) individual users whose tweets may be incorrectly labeled, (ii) groups that use non-standard or informal language more frequently, and (iii) people who rely on sentiment analysis results for monitoring or analysis. The potential negative impact includes unfair interpretation of users’ opinions and misleading sentiment statistics. The severity of this risk is moderate to high if the model is used in real-world applications, and the likelihood is moderate because sentiment labels are subjective and language usage varies widely.

For example, the tweet “that concert was sick (fire emoji) ” expresses positive sentiment, but the word “sick” may cause the model to predict a negative label if it has mostly seen this word in negative contexts during training. Similarly, tweets containing identity-related words or reclaimed language may be misclassified if these terms are correlated with negative sentiment in the training data. Such errors can affect certain groups more than others and can distort overall sentiment analysis results.

We evaluated this risk by manually inspecting misclassified tweets from the validation set, with a focus on examples containing slang, or informal expressions. To reduce potential harm, we clearly state the limitations of our model, avoid suggesting its use for automated moderation or decision-making, and emphasize that overall accuracy may hide different error patterns across language styles. A remaining limitation is the lack of demographic information in the dataset, which makes it difficult to perform a detailed fairness analysis. This model is developed for academic purposes only and is not intended for deployment in real-world systems.

5 Results and Conclusion

This project examined how different neural architectures behave on a tweet sentiment classification task when both model complexity and dataset size are varied. The experiments show that access to more data generally leads to better performance, but only when the model is capable of exploiting it. Non-sequential approaches such as MLPs quickly reach a performance ceiling, whereas models that account for word order and local context perform substantially better. In particular, CNNs stand out as a robust solution, reaching accuracy close to that of LSTMs while remaining simpler to train and easier to scale. The transformer-based encoder did not provide clear gains in this setting, largely due to practical limitations on model depth and training time. Overall, the results suggest that for short and noisy text like tweets, efficient convolutional models remain a highly competitive choice when trained on large datasets

7 Appendix

6 Bibliography

Références

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